



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Business Systems Academic Year 2022– 2023(Odd Semester)

Degree, Semester & Branch: B.Tech – I Semester CSBS

Course Code & Title: GE3151 Problem Solving and Python Programming

Name of the Faculty member (s): Dr.S.Erana Veerappa Dinesh, AP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit IV / Dictionaries-Operations
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO11
- **Activity Chosen:** Crossword Puzzle
- **Justification:**
Various operations of dictionary in python can be easily paired with the help of crossword puzzle which leads to the greater retention of the data among the students.
- **Time Allotted for the Activity:** 15 Minutes
- **Details of the Implementation:**
 - A crossword puzzle is a fun word game that encourages students to remember the idea and try to figure out the solution on their own. Instead of administering standard MCQ tests, it sparks pupils' curiosity.
 - The activity was conducted in a classroom for a duration of 10 minutes. The students were given 12 questions as a crossword puzzle in the topic of dictionaries. They have to think and write the answers in the given puzzle.
 - The activity was conducted during the last 15 minutes of the class after that the answers were corrected. Sample answers written by Ms.B.Krittika and Mr.T.Kishor of I CSBS were shown in figure 1 and figure 2 respectively. The answers for the wrong questions were discussed at the end of the class to make them to know the answer of all the questions.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO2	3	3	2	1	2	2	1	2	1	1

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PSO1
	3	3	2	2	2
Justification for correlation	The students apply the knowledge of dictionary in python	The students analyze the given questions and identify solutions	The students find the solutions for the given question.	The students demonstrate problem-solving skills as a team.	The students apply the operations of dictionaries in python for solving the novel problems

• **Images / Screenshot of the practice:**

KRITTIKA · B
CSBS
22CB027

GE3151 Problem Solving and Python Programming
Unit IV – Lists, Tuples, Dictionaries
Innovative Practice Session- Crossword Puzzle
Dictionaries: Operations and Methods

ACROSS

2. Deletes all entries in the dictionary

5. Brackets are used along with key for access values

6. Checks whether a given key is present in dictionary

7. Another name of dictionary

10. Function is used to sort the keys in dictionary

11. Used for separate the items

DOWN

1. Separated from its values by a colon

2. Returns a shallow copy of the dictionary

3. Iterate through items in dictionary

4. Returns a list of values in dictionary

8. Method removes an item from dictionary

9. Returns a list of tuples in dictionary

Fig 1 : Puzzle solved by Ms. B.Krittika

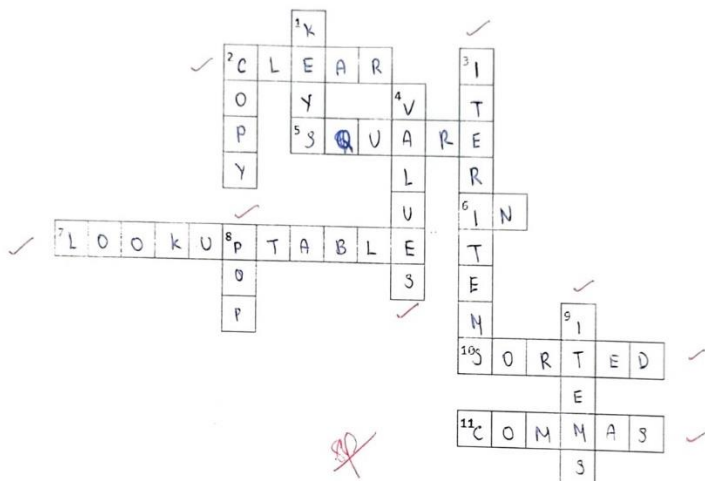
KISHOR T
22CB024

GE3151 Problem Solving and Python Programming

Unit IV – Lists, Tuples, Dictionaries

Innovative Practice Session- Crossword Puzzle

Dictionaries: Operations and Methods



ACROSS

2. Deletes all entries in the dictionary
5. Brackets are used along with key for access values
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1. Separated from its values by a colon
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3. Iterate through items in dictionary
4. Returns a list of values in dictionary
8. Method removes an item from dictionary
9. Returns a list of tuples in dictionary

Fig 2 : Puzzle solved by Mr. T.Kishor

❖ Reflective Critique:

Feedback of practice from students and other stakeholders:

- Some of the students find very easy to identify the various operations used in dictionary.
- Some of them take more time to complete the puzzle need more clarification on the various classes.

❖ Benefit of the practice:

(E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

It helps the students to think on their own and develop the python programs using dictionary operations. The students level of understanding on this topic gets increased.

❖ ***Challenges faced in implementation:***

- The students who didn't understand the concept clearly were unable to complete it within the allotted time.
- Few of them were struggled to find the answer for the questions.

References:

- ❖ https://www.ritrjpm.ac.in/images/computer-science/2021-2022/BV_CS8602_Cross-Word-Puzzle.pdf
- ❖ https://www.ritrjpm.ac.in/images/computer-science/RV_crossword_BSEC.pdf
- ❖ <http://www.classtools.net/crossword/>
- ❖ <https://www.education.com/worksheet-generator/reading/crossword-puzzle/>

Signature of Faculty Member

HOD